

What is a backup?

- ✓ A backup is a replica/copy of an important data.
- ✓ The reason for creating backup is protect data from getting damage or accidentally getting deleted.
- ✓ Backups are useful in the scenarios like:
 - Hardware failure
 - Accidental deletion
 - Malware attack
 - Data corruption
 - Disaster

Core components of backup:

- ✓ *Backup source*
 - The original data source that needs to be backed up.
 - Ex:
 - File, folder, database, VM, system state, etc.
- ✓ *Backup target*
 - The destination where backup will be stored.
 - Ex:
 - Disk drive, tape drive, cloud
- ✓ *RTO*
 - Recovery Time Objective
 - How quickly you need to restore the data?
- ✓ *RPO*
 - Recovery Point Objective
 - How much data loss you can afford?
 - 1 minute
 - 1 hour
 - 1 day

Types of backups:

- ✓ **Full backup**
 - It basically takes FULL backup of system/data etc.
 - It requires large amount of space to store the backed-up data.
 - Restoration is simple, as everything will be restored.
- ✓ **Incremental backup**
 - It will backup only since last backup was done.
 - Faster and efficient to backup
 - Slower in restoration.
- ✓ **Differential backup**
 - It's the backup since last FULL BACKUP.
 - Faster restore
 - More storage is needed.
- ✓ **Synthetic full backup**
 - It merges incremental backup to simulate a full backup.
- ✓ **Mirror backup**
 - Real-time data duplication (RAID 1)
- ✓ **Snapshot backup**
 - Capturing the state of a VM.

Backup design pattern – The 3-2-1 rule

- ✓ **3 copies** of data will be created
 - 1 Primary backup
 - 2 backup copies
- ✓ **2 different media types** for taking backup
 - Disk drives
 - Tape drives
 - Cloud storage
- ✓ **1 copy off-site** must be stored
 - Cloud or
 - Remote location

Backup policies:

- ✓ A backup policy is a rule that helps in:
 - How to create a backup?
 - How to store a backup?
 - How it has to be managed?
- ✓ Components of a backup policy:
 - **Scope** – What to back up?
 - File, folder, system → what to include and what to exclude?
 - **Frequency** – how frequently you want to back up?
 - Daily, hourly, weekly, real-time
 - **Retention** – how long you want to keep the backed-up data?
 - 7 days, 15 days, 30 days, etc...
 - **Storage** – where to store/take the backup?
 - On-prem, cloud, hybrid
 - **Encryption** – data must be encrypted.
 - At rest – when data is stored.
 - In-transit – when data is moving.
 - Or both.
 - **Authentication and Access**
 - Who can manage the data
 - **Compliance** – to ensure to meet the regulatory needs.
 - *HIPAA* – Health Insurance Portability and Accountability Act
 - *GDPR* – General Data Protection Regulation
 - *PCI-DSS* – Payment Card Industry Data Security Standard
 - *SOX* – Sarbanes-Oxley Act

Types of storages for backup:

✓ **Tape drives**

- Tape drives use magnetic tapes.
- It's a low-cost solution
- It can store data for years without requiring any power.
- It must be handled manually.
- Needs to be protected from physical damage.

✓ **Disk storage**

- We can use HDD, SSDs, disk array (NAS, SAN).
- Faster, easy to monitor.
- Costlier than tape drives
- Hardware failure, data corruption over the time.

✓ **Cloud Storage**

- AWS, Azure, GCP
- Quickly accessible, easily scalable, no hardware needed.
- More data means more money.
- Dependent on internet.

How to choose the right back up?

- ✓ Capacity
- ✓ Performance
- ✓ Cost
- ✓ Durability
- ✓ Compliance
- ✓ Compatibility
- ✓ RPO/RTO
- ✓ Security

Backup software and tools:

Windows

- ✓ Using “file history”
- ✓ Using system image backup
- ✓ PowerShell scripts
- ✓ Backup to the cloud
- ✓ On windows server, install a role “Windows Server Backup”

Linux

- ✓ Copy “cp” command
- ✓ TAR for archiving the directory
- ✓ Rsync utility
- ✓ Shell Script
- ✓ Using “dd” command
- ✓ Using SCP command

Enterprise-level backup solution:

- ✓ Veeam backup
 - Standard for virtual, physical and cloud backups.
 - It supports multi-cloud/hybrid environment setup.
- ✓ Commvault
 - Used for on-premises, cloud or SAAS applications
 - Automation and scalability are supported.
- ✓ Rubix
 - Automation-focused and quick restore.
 - Used in hybrid environments
- ✓ Cohesity
 - High automation, instant recovery.
 - Centralized management for enterprise.
- ✓ Acronis cyber protect
 - Integrated backup with cybersecurity features.
- ✓ Veritas Backup
 - Used for apps (new or legacy)

Cloud backup solutions:

- ✓ Back blaze
- ✓ IDrive
- ✓ Dropbox backup
- ✓ Carbonite

Open-source tools:

- ✓ Fbackup
- ✓ Backup PC
- ✓ EaseUS to do backup.

Backup and Recovery procedures:

Backup procedure:

1. Policy and Planning

- Defining the scope – what to be backed up? Data, apps, etc.
- Backup policy – frequency, retention period, media type and access.
- Redundancy – apply 3-2-1 rule.

2. Execution

- Automation – automate the backup procedure.
- Monitoring – use some monitoring tool to get the alerts for success, failed or any warning.
- Security – in-transit or at-rest encryption.

3. Maintenance

- Regularly testing before you actually restore it.
- Documentation – detailed documents must be maintained for knowledge transfer.

Recovery Procedure:

1. Preparation

- Recovery objectives – RTO/RPO
- Prioritization – giving the priority to the critical components first.

2. Restoration steps:

- Detection – identify the data loss.
- Evaluation – scope of the affected data.
- Initialize restore:
 - Retrieve backup media
 - Use the appropriate software for the restoration
 - Monitoring the restoration process.
- Validation – check the integrity of the restored data. And repeat if any error occurs.

3. Post-Recovery

- System hardening – apply patches, RCA
- Testing and feedback – post-incident review.

Monitoring:

Pre-requisites for lab:

- ✓ 1 CentOS 9 server (for Nagios)
- ✓ 1 Windows machine (Server/Client/DC) – for monitoring.
- ✓ Note:
 - Both must ping each other. Use the NAT adapter.
 - Both must ping to internet (google.com).

Monitoring

- ✓ Monitoring is a continuous process of observing health, performance and usage of a system.
- ✓ With monitoring basically, we can collect data like:
 - CPU usage
 - Memory usage
 - Disk usage
 - Uptime
 - Error logs
 - Network logs
 - Application logs
 - Temperature of the hardware device
 - Services
 - We can detect any problems.
 - Alerts
 - RCA
- ✓ Advantages of monitoring:
 - Early alert
 - Performance optimization
 - Improves security
 - Audit and compliance
 - Reduced downtime
- ✓ Disadvantages of monitoring:
 - Complex to setup
 - Cost of the tool
 - Privacy risk
- ✓ Use cases:
 - Server runs out of disk space.
 - Application(s) running slow.
 - High CPU utilization.
 - High memory utilization.

Working of a monitoring tool:

1. Data collection

- ✓ The monitoring tool collects data from different sources:
 - Agent-based method – an agent/software is required to collect the data.
 - Agentless method – system uses SSH, WMI or API to collect the information.
 - Cloud APIs – for Azure, AWS, GCP.

2. Data transmission

- ✓ Then the collected data is sent to the monitoring server at regular interval (few mins, hours, days).
- ✓ It uses various protocols to send the information like HTTP, HTTPS, TCP, UDP.

3. Data processing

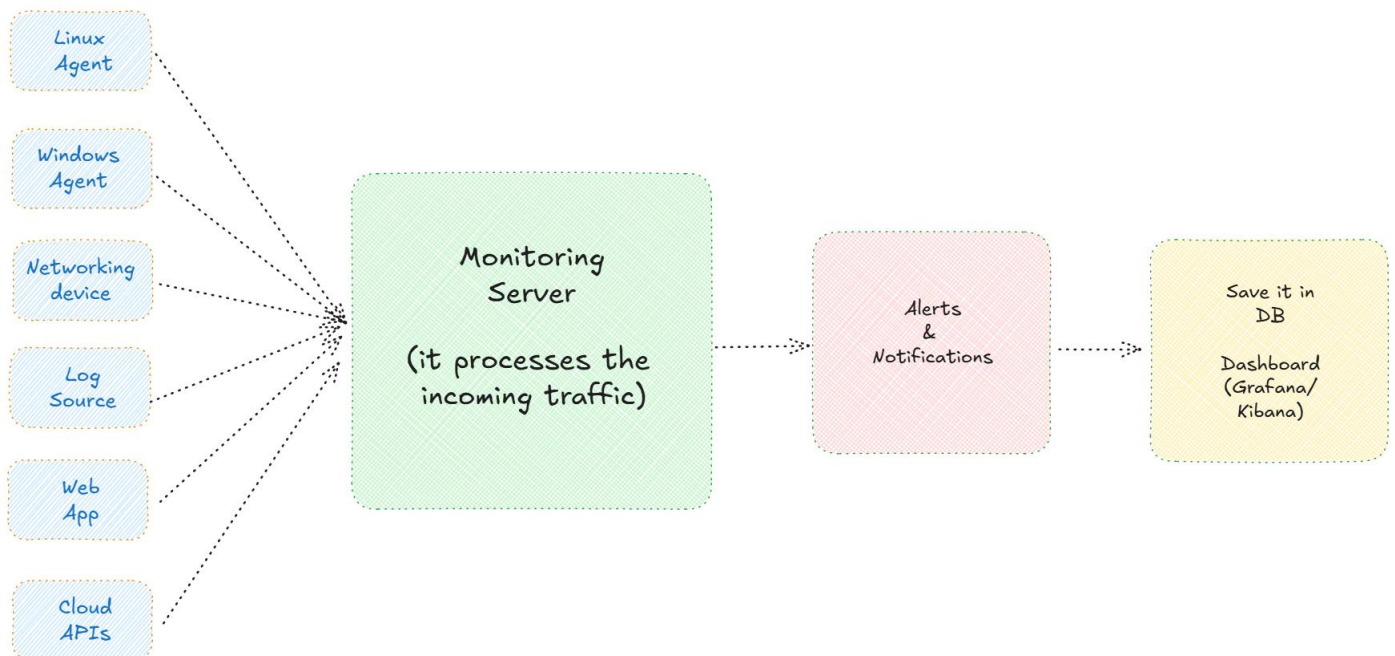
- ✓ The monitoring tool then processes the incoming data.
- ✓ Like:
 - Check the CPU threshold.
 - Applies rules to the increased threshold.
 - Stores the data in the tool's database.

4. Alerts and notifications

- ✓ If any issues are detected then monitoring tool will trigger alert using:
 - Email, SMS, Teams message calls, etc.

5. Visualization and reporting

- ✓ Monitoring tool will show the report in real-time.
- ✓ By using:
 - Graphs
 - Charts
 - Heatmaps
 - Logs
 - Custom dashboards (Grafana, Kibana (ELK - Elasticsearch, Logstash, and Kibana)).



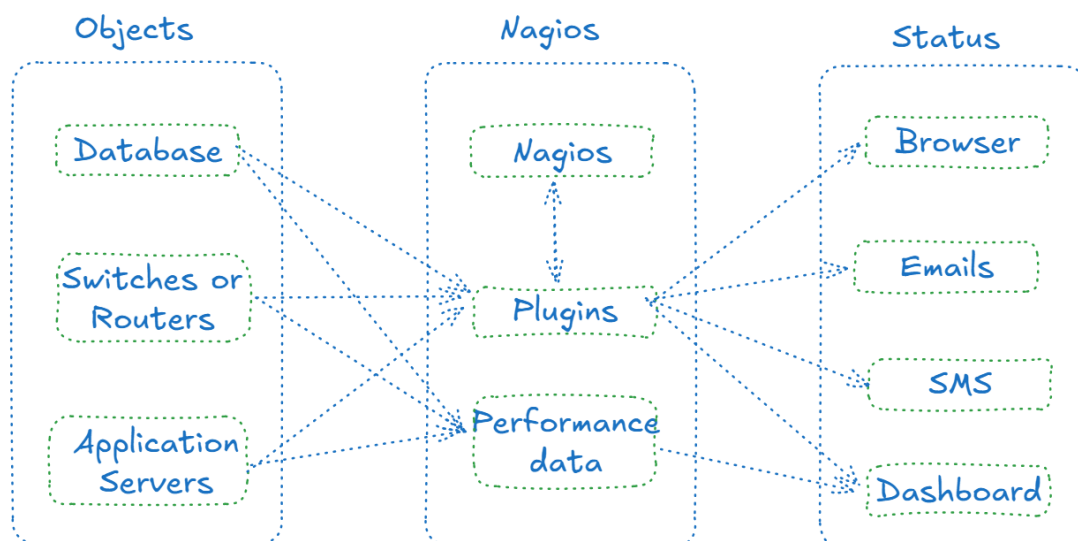
List of Monitoring tools:

Categories of Monitoring tools:

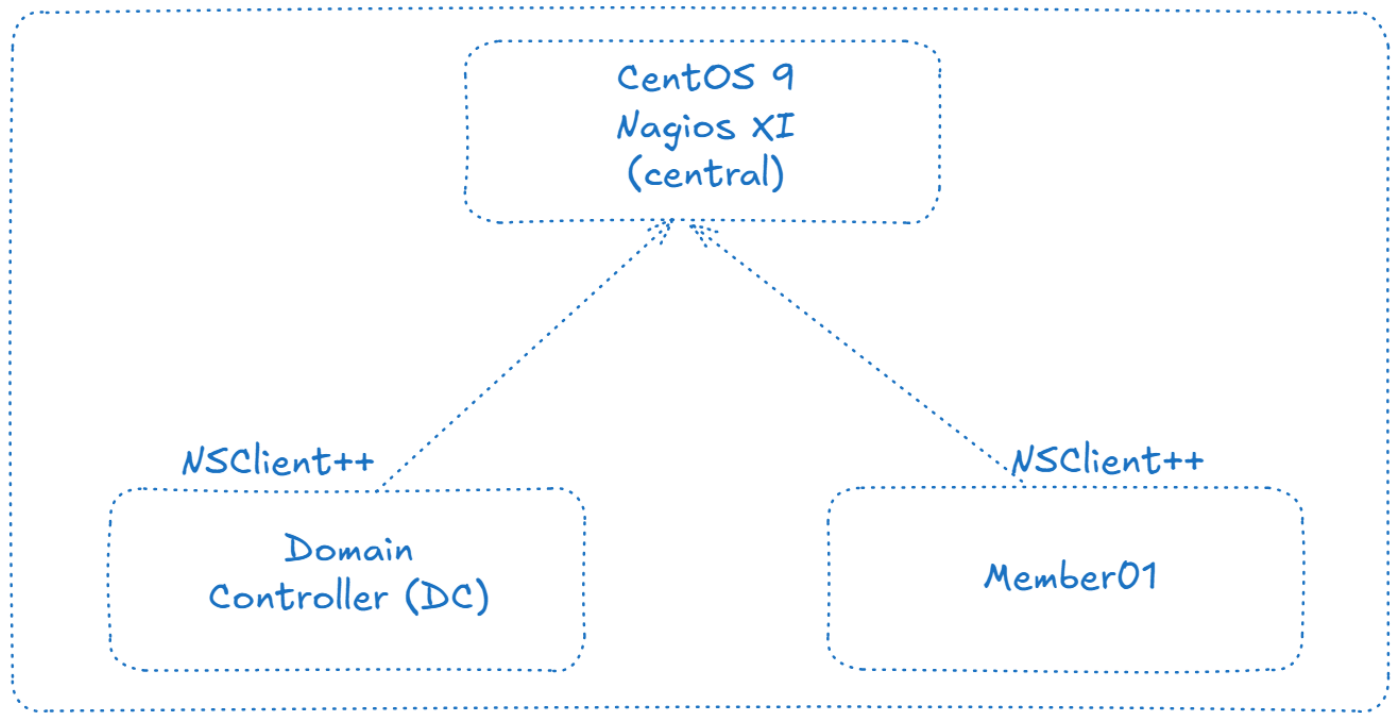
1. Free/Open-source tools
 - a. **Nagios core** – CLI tool with customizations.
 - b. **Zabbix** – open-source tool with dashboard.
 - c. **Prometheus** – a free tool integrated with Grafana.
 - d. **LibreNMS** – networking-focused monitoring tool, specially for networking devices.
 - e. **Sensu core** – for monitoring pipeline in DevOps.
2. Enterprise/Paid tools
 - a. **Nagios XI** – a production-level web-tool with dashboard.
 - b. **Zabbix Enterprise support** – Enterprise-level monitoring tool with dashboard.
 - c. **SolarWinds Network performance monitoring (NPM)** – used for network monitoring.
 - d. **DataDog** – full stack cloud monitoring tool.
 - e. **Dynatrace** – AI-Driven tool
 - f. **New Relic** – enterprise-level monitoring tool.
 - g. **Splunk** – real-time metrics log and tracing tool.
3. Cloud Monitoring tools
 - a. **Amazon CloudWatch** – monitoring instance, services and logs.
 - b. **Azure Monitor** – monitoring solution for resources in azure.
 - c. **Google cloud operations** – Native monitoring tool for google resource.
 - d. **Grafana cloud** – hosted with Prometheus and Loki.
 - e. **ELK** – Elasticsearch, Logstash and Kibana.
 - f. **Sentry** – for frontend and backend code.

Nagios

- ✓ Nagios is a powerful open-source monitoring tool.
- ✓ It helps in monitoring:
 - Systems (Linux, Windows)
 - Network
 - Applications
 - Services
 - Disk drives
 - Processes
 - Hardware temperature
 - State of the system
- ✓ Nagios is available in 2 modes:
 - Nagios Core
 - It's a free tool.
 - It completely works in CLI.
 - It is highly customizable.
 - Nagios XI
 - It's free for 60 days.
 - All the mgmt related work is done using web console (GUI).
 - In the enterprise version, support is also available.
- ✓ Nagios architecture:
 - **Nagios Core**
 - Central engine to execute the commands.
 - **Plug-ins**
 - Scripts/binaries to work with the Nagios.
 - Check_ping
 - Check_disk
 - **NRPE (Nagios Remote Plugin Executor)**
 - This monitors the linux/UNIX based systems.
 - **NSCP (NS Client ++)**
 - This monitors the windows machines.
 - **Web interface**
 - GUI to visualize the reports, logs, notifications.
 - **Event handler**
 - Scripts that triggered when alert occurs.
 - **Notification system**
 - Sends alerts via email, SMS, call, push notifications.



Lab setup:




```
Country Name (2 letter code) [XX]:IN
State or Province Name (full name) []:MP
Locality Name (eg, city) [Default City]:BPL
Organization Name (eg, company) [Default Company Ltd]:Self
Organizational Unit Name (eg, section) []:Trainer
Common Name (eg, your name or your server's hostname) []:server
Email Address []:

[root@server ~]#
[root@server ~]# chmod 600 /usr/local/nagiosxi/var/certs/nagios
[root@server ~]# chmod 644 /usr/local/nagiosxi/var/certs/nagios
[root@server ~]# systemctl restart httpd
[root@server ~]#
```